



On-Premise Azure Pipelines Implementation Guide For Java Applications

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DEVELOPER API GUIDE DOCUMENT

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Revision History

Date	Version	Author	Comments
10.01.2025	v1.0.0	Kennedy N. Gatimu	Initial document creation

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1.0 Introduction

This guide provides step-by-step instructions for implementing Azure DevOps CI/CD pipelines for on-premise Java applications. The goal is to streamline deployment, ensure consistency, and enhance security by managing sensitive configuration data via Azure Key Vault.

The example application used in this guide is **tech-b2c-business-cd-ms**, and the configurations will be tailored to match its requirements.

I have attached sample [azure-pipelines.yml](#) and [.gitignore](#) files. Customize them as per your application's requirements.

2.0 Clone Pipelines Repository

Clone the Azure DevOps pipelines repository from the following URL:

```
git clone https://dev.azure.com/ICT-DevOps/Transaction-Processors-GAG/_git/pipelines
```

3.0 Repository Directory Structure

The cloned repository has the following directory structure:

```
pipelines
├── apm
├── deploy
│   ├── prod
│   └── uat
├── java_env
│   ├── prod
│   └── uat
├── scripts
├── tanuki
│   ├── prod
│   │   ├── bin
│   │   ├── conf
│   │   └── lib
│   └── uat
│       ├── bin
│       ├── conf
│       └── lib
```

4.0 Pipelines Directory Description

4.0.1 APM

Holds the **elastic-apm-agent-1.51.0.jar** for application monitoring using **ELK**.

4.0.2 Deploy

Contains **Finacle ISO-related** configurations for UAT and PROD.

Key files:

- drc_channel.xml
- drc_mux.xml
- startPBQB.xml

4.0.3 Java_env

Stores two configuration files for both UAT and PROD environments

- **env.sh**: A bash script that stores all environment variables needed for the application to run. These variables are stored in Azure Key Vault and must be replaced during the build pipeline.
- **j_env.sh**: A bash script that defines the Java path

4.0.4 Scripts

Contains two important bash script files:

- **azure-pipelines-helper.sh**: Used by the azure-pipelines.yml file to perform tasks during the build (CI) pipeline, such as creating a common deployment directory, copying necessary files, and archiving the deployment directory.
- **application-deployment.sh**: Used during the release pipeline (CD) to create required directories, copy deployment artifacts, set up the environment, and start the application.

4.0.5 Tanuki

Holds all Tanuki-related files for both UAT and PROD environments. It includes the following subdirectories:

- **bin**: Contains the application start script (**tech-b2c-business-cd-ms.sh**) and the Tanuki wrapper file (**wrapper**).
- **conf**: Contains the application's Tanuki configuration file (**tech-b2c-business-cd-ms.conf**).
- **lib**: Contains Tanuki library files (**libwrapper.so** and **wrapper.jar**).

5.0 Copy Directory to Application Root

Copy the cloned pipelines directory to the root of your Java application (tech-b2c-business-cd-ms).

6.0 Edit Configuration Files

Edit the following directories and files to match your application.

For this guide we are using the **tech-b2c-business-cd-ms** application.

6.0.1 APM

No changes required (Elastic APM version remains 1.51.0).

6.0.2 Deploy

Update the **mux** configurations in the UAT and PROD directories to match the respective **ISO Finacle** instance of the respective bank.

6.0.3 Java_env

Update **env.sh** and **j_env.sh** files in the UAT and PROD directories to match the desired configurations and Java path.

- **env.sh:** Set environment-specific variables.
- **j_env.sh:** Update Java path.

6.0.4 Scripts

Caution: Do not update these bash scripts manually.

6.0.5 Tanuki

a. Bin directory

1. Rename the name of the tech-b2c-business-cd-ms.sh file to match your application name (e.g., tech-b2c-business-cd-ms.sh).
2. Update the following lines in the tech-b2c-business-cd-ms.sh file to match your application:

```
APP_NAME=tech-b2c-business-cd-ms
APP_LONG_NAME=tech-b2c-business-cd-ms
WRAPPER_CMD="./wrapper"
WRAPPER_CONF="../conf/tech-b2c-business-cd-ms.conf"
```

b. Conf directory

1. Rename the .conf file to match the application name (e.g., tech-b2c-business-cd-ms.conf).
2. Update the following lines in the .conf file to match your application configs:

```

wrapper.java.command=/u01/gag/sw/jdk-17.0.6/bin/java
set.MYAPP_NAME=tech-b2c-business-cd-ms
set.MYAPP_HOME=/u01/gag/drcb2c/tech-b2c-business-cd-ms
wrapper.java.classpath.1=%MYAPP_HOME%/lib/wrapper.jar
wrapper.java.classpath.2=%MYAPP_HOME%/bin/tech-b2c-business-cd-ms-0.0.1-SNAPSHOT.jar
wrapper.java.classpath.3=%MYAPP_HOME%/bin/lib/*.jar
wrapper.app.parameter.1=com.equitybank.gag.telco_b2c.TelcoB2cMsApplication

```

3. Modify JVM settings only if necessary:
Caution! Update these settings, only if you understand how JVM works.
 Otherwise **DO NOT** change the settings.

```

wrapper.java.additional.auto_bits=TRUE
wrapper.java.additional.1=-Xms512m
wrapper.java.additional.2=-Xmx3072m
wrapper.java.additional.3=-XX:NewRatio=3

```

4. **Optional:** If using ELK APM Agent, update:

```

wrapper.java.additional.1=-javaagent:%MYAPP_HOME%/bin/apm/elastic-apm-agent-1.51.0.jar
wrapper.java.additional.2=-Delastic.apm.service_name=tech-b2c-business-cd-ms
wrapper.java.additional.3=-Delastic.apm.secret_token=WNqo7qepTnVicOF4W1T72+RqbOeDboFRu0MjEhPIjvw=
wrapper.java.additional.4=-Delastic.apm.server_url=http://10.11.6.229:8200
wrapper.java.additional.5=-Delastic.apm.environment=develop
wrapper.java.additional.6=-Delastic.apm.application_packages=com.equitybank.gag.telco_b2c

```

c. Lib directory

No modifications required.

7.0 Update Application Root Files

7.0.1 *azure-pipelines.yml*

Update the following variables to match your application:

```
JAVA_VERSION: '1.17'
APP_NAME: 'tech-b2c-business-cd-ms-0.0.1-SNAPSHOT.jar'

stages:
- stage: BuildUAT
  condition: and(always(), contains(variables['Build.SourceBranch'], 'refs/heads/release'))
  displayName: UAT
  variables:
    ENVIRONMENT: 'uat'

- stage: BuildPROD
  condition: and(always(), contains(variables['Build.SourceBranch'], 'refs/heads/main'))
  displayName: PROD
  variables:
    ENVIRONMENT: 'prod'
```

7.0.2 *.gitignore*

Add the following rules:

```
#####
# Allow jar files in the pipelines
#####
!pipelines/**/*.jar
!pipelines/templates/.gitignore
```

8.0 Sample azure-pipelines.yml and .gitignore Files

8.0.1 *azure-pipelines.yml*

```
trigger:
- release
- main

variables:
  tag: '$(Build.BuildId)'
  vmImageName: 'ubuntu-latest'
  azureSubscription: 'Enterprise Core'
  uatKeyVaultName: 'kv-uat-omnichannel'
  prodKeyVaultName: 'kv-preprod-omnichannel'

# Change these values as required
JAVA_VERSION: '1.17'
```

```

APP_NAME: 'tech-b2c-business-cd-ms-0.0.1-SNAPSHOT.jar'

stages:
- stage: BuildUAT
  condition: and(always(), contains(variables['Build.SourceBranch'], 'refs/heads/release'))
  displayName: UAT
  variables:
    ENVIRONMENT: 'uat'
  jobs:
  - job: Build
    displayName: Build
    pool:
      vmImage: $(vmImageName)
    steps:
    - task: AzureKeyVault@2
      displayName: 'Connect To Azure UAT KeyVault'
      inputs:
        azureSubscription: $(azureSubscription)
        KeyVaultName: $(uatKeyVaultName)
        SecretsFilter: '*'
        RunAsPreJob: false

    - task: replacetokens@5
      displayName: 'Replace Azure KeyVault Tokens'
      inputs:
        folderPath: '$(System.DefaultWorkingDirectory)'
        targetFiles: |
          **/*uat.yml
          **/*uat.yaml
          **/*uat.properties
          pipelines/java_env/uat/*.sh
        encoding: 'auto'
        writeBOM: true
        actionOnMissing: 'warn'
        keepToken: false
        tokenPrefix: '#{ '
        tokenSuffix: ' }#'
        useLegacyPattern: false
        enableTransforms: false
        enableTelemetry: true

    - script: 'rm -rf ~/.m2/repository'
      displayName: 'Delete Maven Local Repository'

    # This task is optional. Comment if not needed
    - task: Maven@4
      displayName: 'Maven Install ojdbc dependency'
      inputs:
        mavenPomFile: 'pom.xml'
        mavenOptions: '-Xmx3072m'
        goals: 'install:install-file'
        options: '-Dfile="ojdbc6-11.2.0.3.jar" -DgroupId="com.oracle" -DartifactId="ojdbc6"
-Dversion="11.2.0.3" -Dpackaging=jar'
        javaHomeOption: 'JDKVersion'
        jdkVersionOption: $(JAVA_VERSION)
        mavenVersionOption: 'Default'
        mavenAuthenticateFeed: false

```

```

    effectivePomSkip: false
    sonarQubeRunAnalysis: false

  - task: Maven@4
    displayName: 'Maven Clean Install'
    inputs:
      mavenPomFile: 'pom.xml'
      options: '-Dspring.profiles.active=uat -DskipTests'
      javaHomeOption: 'JDKVersion'
      jdkVersionOption: $(JAVA_VERSION)
      mavenVersionOption: 'Default'
      publishJUnitResults: false
      mavenAuthenticateFeed: false
      effectivePomSkip: false
      sonarQubeRunAnalysis: false
      goals: clean install

  - task: Bash@3
    displayName: Running Azure-Pipelines-Helper Script
    inputs:
      targetType: 'filePath'
      filePath: 'pipelines/scripts/azure-pipelines-helper.sh'
    env:
      APP_NAME: $(APP_NAME)
      ENVIRONMENT: $(ENVIRONMENT)
      BUILD_ARTIFACTS_STAGING_DIR: $(Build.ArtifactStagingDirectory)

  - task: CopyFiles@2
    displayName: Copy application-deployment.sh To Staging Directory
    inputs:
      SourceFolder: '$(System.DefaultWorkingDirectory)/pipelines/scripts'
      Contents: |
        application-deployment.sh
      TargetFolder: '$(Build.ArtifactStagingDirectory)'

  - task: PublishBuildArtifacts@1
    inputs:
      PathToPublish: '$(Build.ArtifactStagingDirectory)'
      ArtifactName: 'uat_drop'
      publishLocation: 'Container'

- stage: BuildPROD
  condition: and(always(), contains(variables['Build.SourceBranch'], 'refs/heads/main'))
  displayName: PROD
  variables:
    ENVIRONMENT: 'prod'
  jobs:
    - job: Build
      displayName: Build
      pool:
        vmImage: $(vmImageName)
      steps:
        - task: AzureKeyVault@2
          displayName: 'Connect To Azure Prod KeyVault'
          inputs:
            azureSubscription: $(azureSubscription)
            KeyVaultName: $(prodKeyVaultName)

```

```

    SecretsFilter: '*'
    RunAsPreJob: false

- task: replacetokens@5
  displayName: 'Replace Azure KeyVault Tokens'
  inputs:
    folderPath: '$(System.DefaultWorkingDirectory)'
    targetFiles: |
      **/*prod.yml
      **/*prod.yaml
      **/*prod.properties
      pipelines/java_env/prod/*.sh
    encoding: 'auto'
    writeBOM: true
    actionOnMissing: 'warn'
    keepToken: false
    tokenPrefix: '#{ '
    tokenSuffix: ' }#'
    useLegacyPattern: false
    enableTransforms: false
    enableTelemetry: true

- script: 'rm -rf ~/.m2/repository'
  displayName: 'Delete Maven Local Repository'

# This task is optional. Comment if not needed
- task: Maven@4
  displayName: 'Maven Install ojdbc dependency'
  inputs:
    mavenPomFile: 'pom.xml'
    mavenOptions: '-Xmx3072m'
    goals: 'install:install-file'
    options: '-Dfile="ojdbc6-11.2.0.3.jar" -DgroupId="com.oracle" -DartifactId="ojdbc6"
-Dversion="11.2.0.3" -Dpackaging=jar'
    javaHomeOption: 'JDKVersion'
    jdkVersionOption: $(JAVA_VERSION)
    mavenVersionOption: 'Default'
    mavenAuthenticateFeed: false
    effectivePomSkip: false
    sonarQubeRunAnalysis: false

- task: Maven@4
  displayName: 'Maven Clean Install'
  inputs:
    mavenPomFile: 'pom.xml'
    options: '-Dspring.profiles.active=prod -DskipTests'
    javaHomeOption: 'JDKVersion'
    jdkVersionOption: $(JAVA_VERSION)
    mavenVersionOption: 'Default'
    publishJUnitResults: false
    mavenAuthenticateFeed: false
    effectivePomSkip: false
    sonarQubeRunAnalysis: false
    goals: clean install

- task: Bash@3
  displayName: Running Azure-Pipelines-Helper Script

```

```

inputs:
  targetType: 'filePath'
  filePath: 'pipelines/scripts/azure-pipelines-helper.sh'
env:
  APP_NAME: $(APP_NAME)
  ENVIRONMENT: $(ENVIRONMENT)
  BUILD_ARTIFACTS_STAGING_DIR: $(Build.ArtifactStagingDirectory)

- task: CopyFiles@2
  displayName: Copy application-deployment.sh To Staging Directory
  inputs:
    SourceFolder: '$(System.DefaultWorkingDirectory)/pipelines/scripts'
    Contents: |
      application-deployment.sh
    TargetFolder: '$(Build.ArtifactStagingDirectory)'

- task: PublishBuildArtifacts@1
  inputs:
    PathToPublish: '$(Build.ArtifactStagingDirectory)'
    ArtifactName: 'prod_drop'
    publishLocation: 'Container'

```

8.0.2 .gitignore

```

#####
# Project-Specific
#####
/src/main/webapp/content/css/main.css
/target/classes/static/**
/src/test/javascript/coverage/

#####
# Node.js
#####
/node_modules/
/npm-debug.log.*
/.awcache/
/.cache-loader/

#####
# SASS
#####
.sass-cache/

# Maven
/target/
.log/
pom.xml.*

# Gradle
/build/
.gradle/

#####

```

```
# Java
#####
.mtj.tmp/
*.class
*.jar
*.war
*.ear
*.nar
*.db
hs_err_pid*

#####
# IDE Configurations
#####
# IntelliJ
.idea/
/*.iml

# Eclipse
.settings/
/bin/
.metadata
.classpath
.project

# NetBeans
nbproject/private/
build/
dist/

#####
# Visual Studio Code
#####
.vscode/
*.code-workspace

#####
# Operating System Files
#####
.DS_Store
Thumbs.db

#####
# Logs
#####
*.log*

#####
# Code Coverage
#####
/coverage/
/.nyc_output/

#####
# Hidden and Local Files
#####
.gitignore
```

```
.editorconfig
.gitattributes
.lintstagedrc.cjs
.prettierrc
.prettierrc
.yo-rc.json

#####
# Allow jar files in the pipelines
#####
!pipelines/**/*.jar
!pipelines/templates/.gitignore
```

9.0 Edit Release CD Pipeline

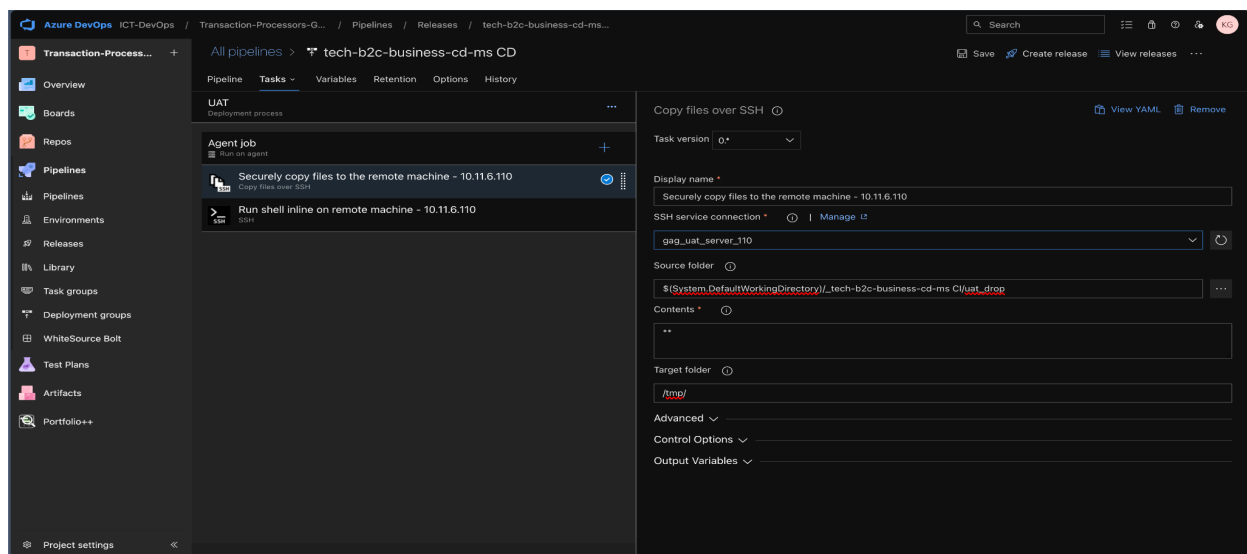
From Azure release pipeline for the specified repository:

- a. Create the release UAT and PROD CD pipeline if it does not exist
- b. On either stage, depending on the environment of choice, create 2 tasks as below;
 - i. Copy files over SSH
 - ii. SSH

9.0.1 Copy files over SSH

Fill in the below form fields with below specified values:

- a. **Display name:** Securely copy files to the remote machine - 10.11.6.110
- b. **SSH service connection:** gag_uat_server_110
- c. **Source folder:** \$(System.DefaultWorkingDirectory)/_tech-b2c-business-cd-ms CI/uat_drop
- d. **Contents:** **
- e. **Target folder:** /tmp/



9.0.2 SSH

Run shell commands or a script on a remote machine using SSH.

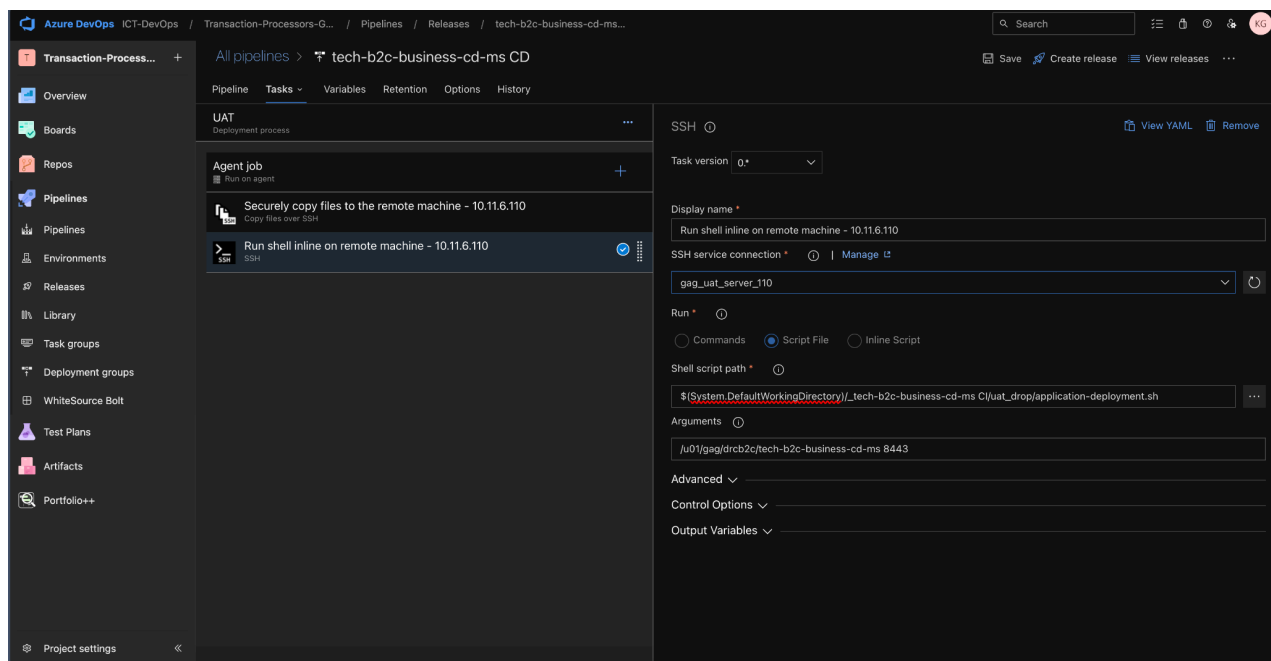
Fill in the below form fields with below specified values:

- Display name:** Run shell inline on remote machine - 10.11.6.110
- SSH service connection:** gag_uat_server_110
- Run:** Script File
- Shell script path:** \$(System.DefaultWorkingDirectory)/_tech-b2c-business-cd-ms CI/uat_drop/application-deployment.sh
- Arguments:** /u01/gag/drcb2c/tech-b2c-business-cd-ms 8443

NOTE:

Arguments field takes two values, <**Application Home Directory Path**> and the <**Application Port**> separated by space.

Example: /u01/gag/drcb2c/tech-b2c-business-cd-ms 8443



10.0 Conclusion

This guide provides a structured approach to implementing Azure DevOps CI/CD pipelines for on-premise Java applications. By following these steps, you can efficiently manage deployments while maintaining security and consistency.